

1. Determine whether the following functions are onto, one to one, both, or neither. Try using your intuition first, and then verify your guess through the definitions.

- $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 2x + 3$.
- $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$.
- $f : [0, \infty) \rightarrow [0, \infty)$ where $f(x) = x^2$
- $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = e^x$.

2. Determine if the function $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ defined by $f(p, q) = \frac{p}{q}$ is onto, one to one, both, or neither.

3. Prove or disprove: if $f \circ g$ is one-to-one then f is one-to-one.

4. Using an argument analogous to Cantor's diagonal argument, show that there does not exist an onto function from a set X into its powerset $\mathcal{P}(X)$. Hint¹

¹For any $f : X \rightarrow \mathcal{P}(X)$ show the set $\{x : x \notin f(x)\}$ is not in the range of f .